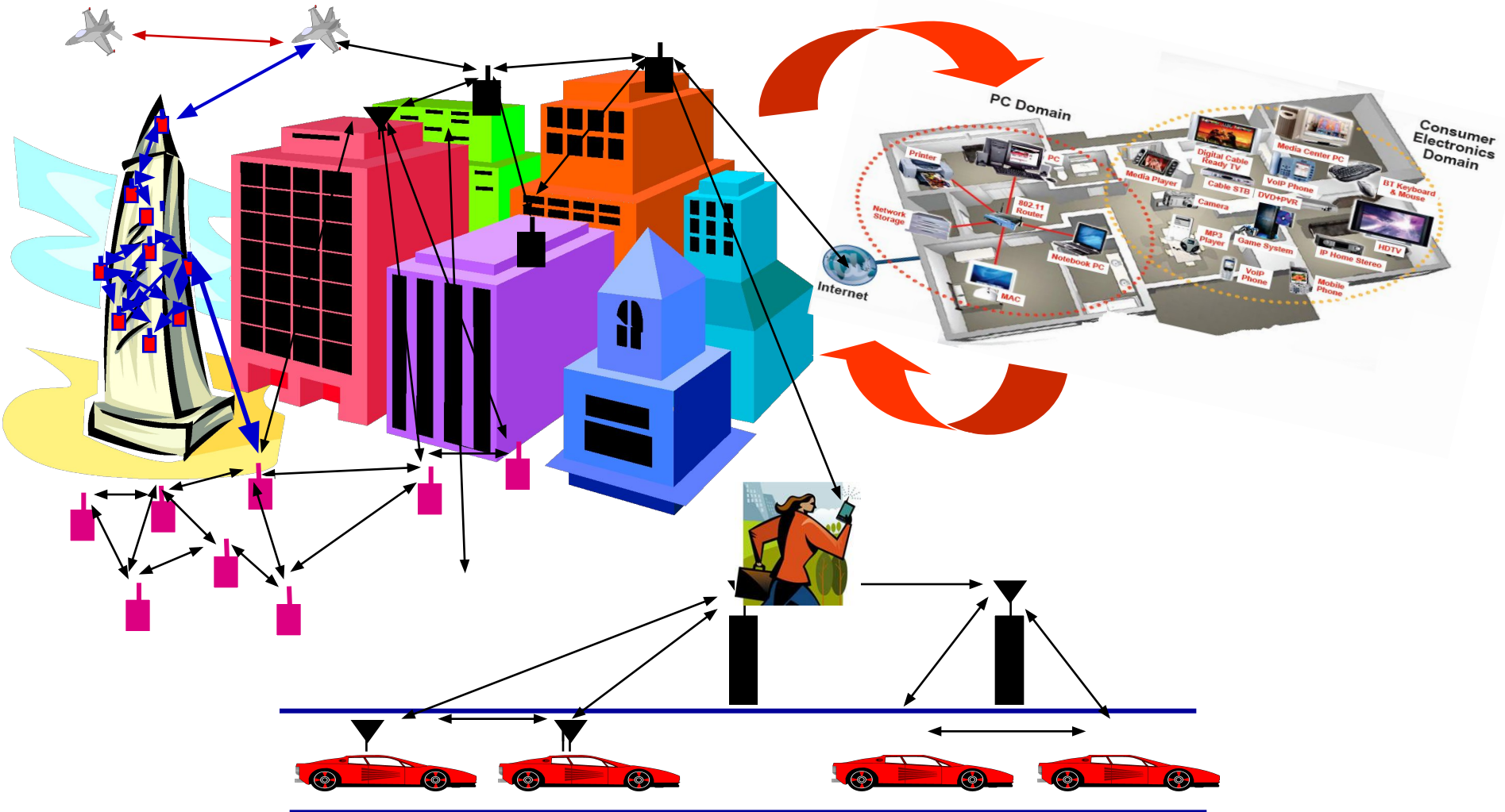


CCE 423: Wireless and Cellular Communications

Prof. Dr. Md. Samsuzzaman



Recommended Book

1. **Mobile Communications --JOCHEN SCHILLER (2nd)**
2. **Wireless Communications --Rappaport**
3. **WIRELESS AND CELLULAR TELECOMMUNICATIONS**
-William C. Y. Lee (3rd)
4. **WIRELESS COMMUNICATIONS – Andrea Goldsmith**

Outline

- Course Basics
- Course Syllabus
- Wireless History
- The Wireless Vision
- Technical Challenges
- Current/Emerging Wireless Systems
- Spectrum Regulation and Standards
- Emerging Wireless Systems

Wireless communications

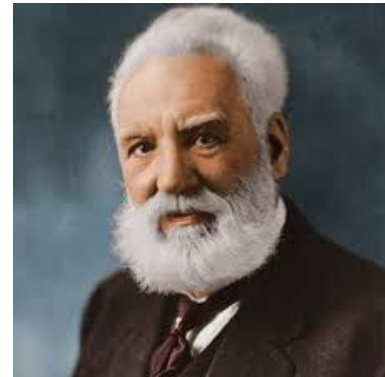
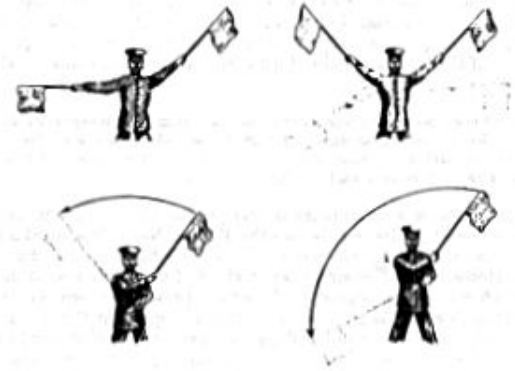
Wireless communications is a type of data communication that is performed and delivered wirelessly. This is a broad term that incorporates all procedures and forms of connecting and communicating between two or more devices using a wireless signal through wireless communication technologies and devices.

Cellular communication

A cellular network or mobile network is a communication network where the last link is wireless. The network is distributed over land areas called cells, each served by at least one fixed-location transceiver, known as a cell site or base station.

Early history of wireless communication

- Many people in history used light for communication
 - heliographs, flags („semaphore“), ...
 - 150 BC smoke signals for communication; (Polybius, Greece)
 - 1794, optical telegraph, Claude Chappe
- Here electromagnetic waves are of special importance:
 - 1831 Faraday demonstrates electromagnetic induction
 - J. Maxwell (1831-79): theory of electromagnetic Fields, wave equations (1864)
 - H. Hertz (1857-94): demonstrates with an experiment the wave character of electrical transmission through space (1886, in Karlsruhe, Germany, at the location of today's University of Karlsruhe)
 - Alexander Graham Bell's invention and marketing of the telephone in 1876



History of wireless communication I

- 1895 Guglielmo Marconi
 - first demonstration of wireless telegraphy (digital!)
 - long wave transmission, high transmission power necessary ($> 200\text{kw}$)
- 1907 Commercial transatlantic connections
 - huge base stations (30 100m high antennas)
- 1915 Wireless voice transmission New York - San Francisco
- 1920 Discovery of short waves by Marconi
 - reflection at the ionosphere
 - smaller sender and receiver, possible due to the invention of the vacuum tube (1906, Lee DeForest and Robert von Lieben)
- 1926 Train-phone on the line Hamburg - Berlin
 - wires parallel to the railroad track



History of wireless communication II

- 1928 many TV broadcast trials (across Atlantic, color TV, TV news)
- 1933 Frequency modulation (E. H. Armstrong)
- 1958 A-Netz in Germany
 - analog, 160MHz, connection setup only from the mobile station, no handover, 80% coverage, 1971 11000 customers
- 1972 B-Netz in Germany
 - analog, 160MHz, connection setup from the fixed network too (but location of the mobile station has to be known)
 - available also in A, NL and LUX, 1979 13000 customer in D
- 1979 NMT at 450MHz (Scandinavian countries)
- 1982 Start of GSM-specification
 - goal: pan-European digital mobile phone system with roaming
- 1983 Start of the American AMPS (Advanced Mobile Phone System, analog)
- 1984 CT-1 standard (Europe) for cordless telephones

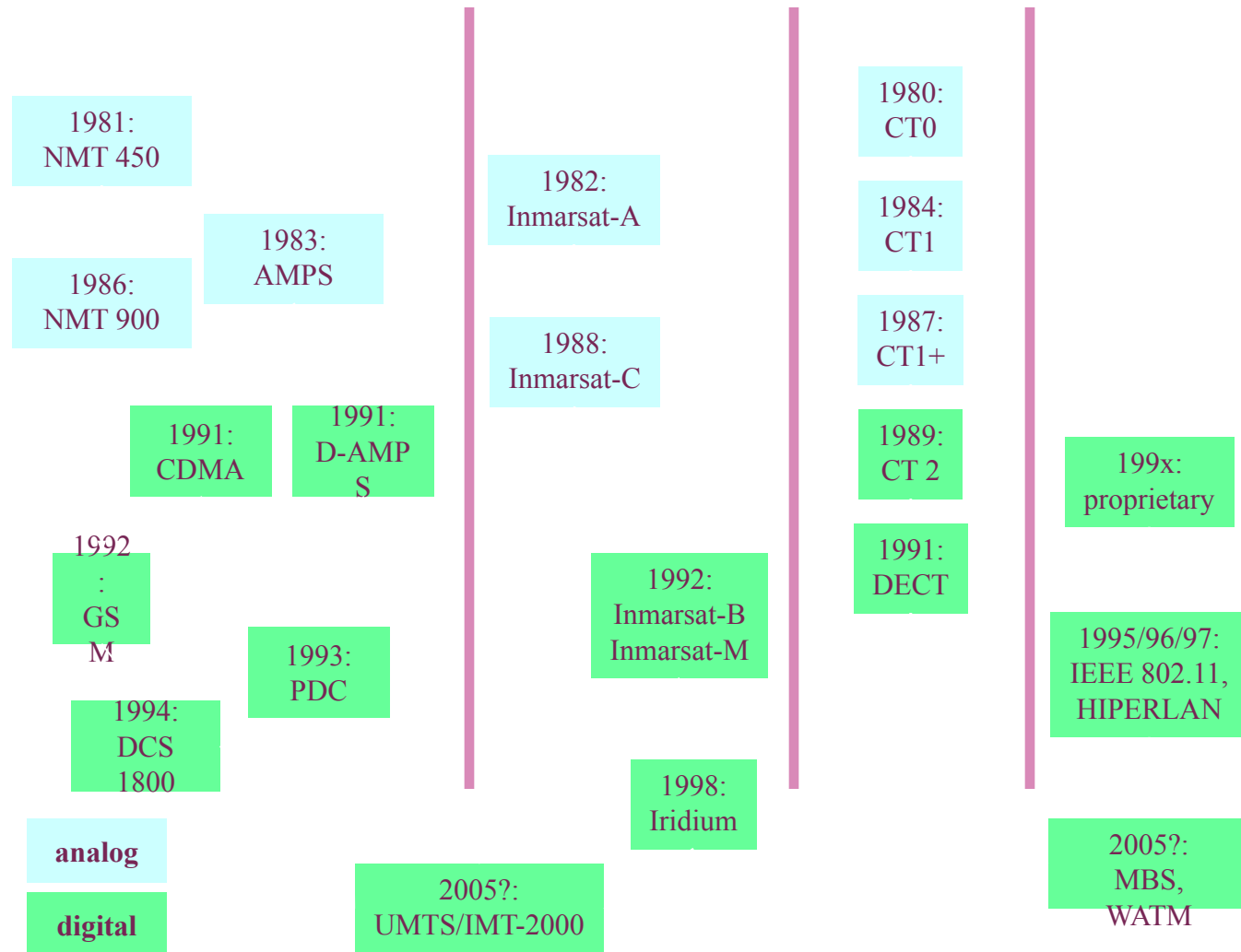
History of wireless communication III

- 1986 C-Netz in Germany
 - analog voice transmission, 450MHz, hand-over possible, digital signaling, automatic location of mobile device
 - still in use today (as T-C-Tel), services: FAX, modem, X.25, e-mail, 98% coverage
- 1991 Specification of DECT
 - Digital European Cordless Telephone (today: Digital Enhanced Cordless Telecommunications)
 - 1880-1900MHz, ~100-500m range, 120 duplex channels, 1.2Mbit/s data transmission, voice encryption, authentication, up to several 10000 user/km², used in more than 40 countries
- 1992 Start of GSM (Global System for Mobile communication)
 - in D as D1 and D2, fully digital, 900MHz, 124 channels
 - automatic location, hand-over, cellular
 - roaming in Europe - now worldwide in more than 100 countries
 - services: data with 9.6kbit/s, FAX, voice, ...

History of wireless communication IV

- 1996 - HiperLAN (High Performance Radio Local Area Network)
 - ETSI, standardization of type 1: 5.15 - 5.30GHz, 23.5Mbit/s
 - recommendations for type 2 and 3 (both 5GHz) and 4 (17GHz) as wireless ATM-networks (up to 155Mbit/s)
- 1997 - Wireless LANs
 - many products with proprietary extensions out there already
 - IEEE-Standard, 2.4 - 2.5GHz, 2Mbit/s
- 1998 - Specification of GSM successors
 - UMTS (Universal Mobile Telecommunication System) as the European proposal for IMT-2000
- 1998 - Specification for next generation CDMA starts
 - Qualcomm starts work on wideband CDMA spec.
- 1999 - Specification of IEEE802.11b
 - increased BW to 11Mbit/s
- 2000 - Bluetooth Specification
 - 1Mbit/s specification, single cell
 - Work on 10Mbit/s spec. with multi cell capability initiated

Wireless systems: overview of the development



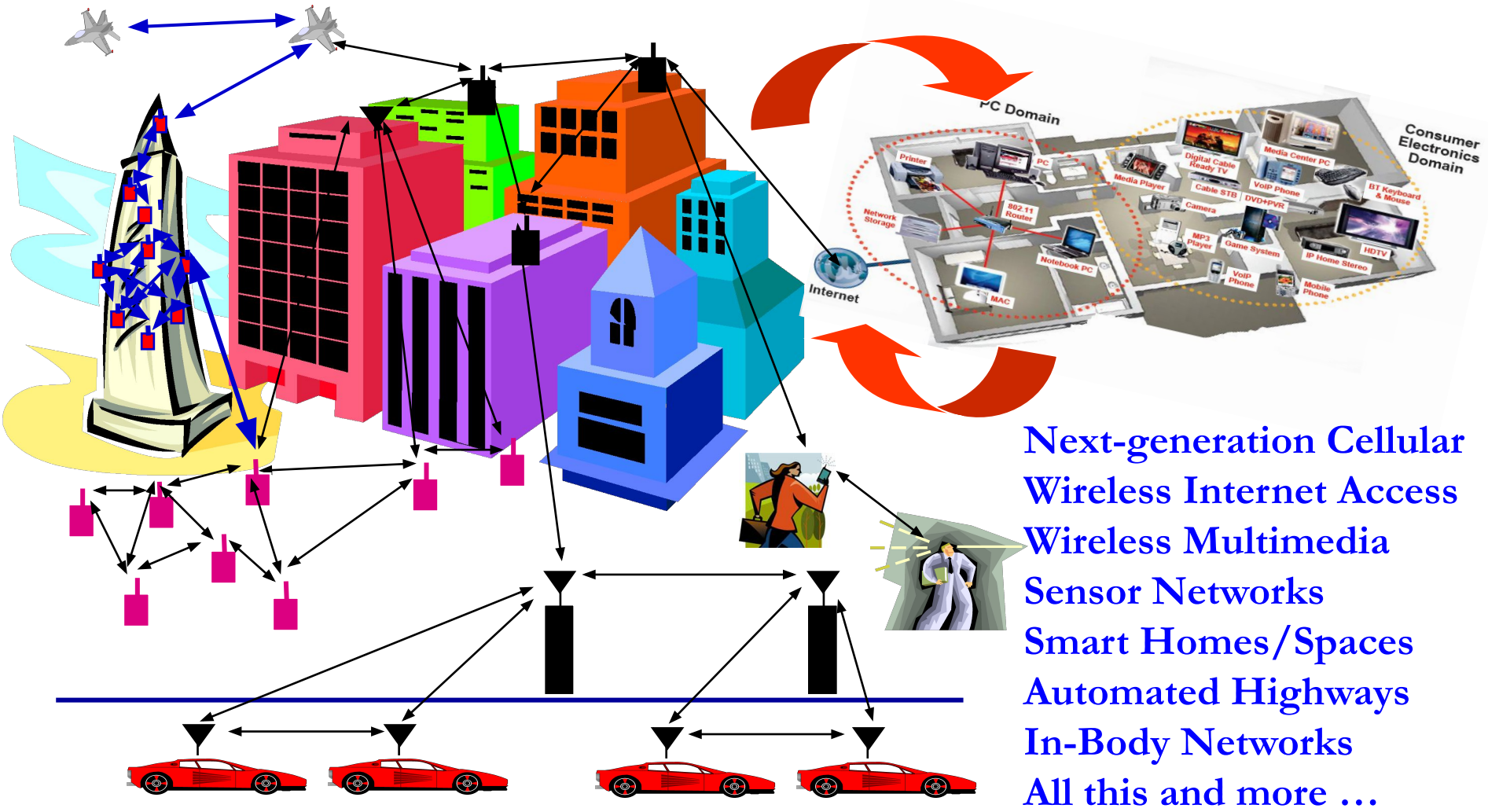
The future: ITU-R - Recommendations for IMT-2000

- M.687-2
 - IMT-2000 concepts and goals
 - M.816-1
 - framework for services
 - M.817
 - IMT-2000 network architectures
 - M.818-1
 - satellites in IMT-2000
 - M.819-2
 - IMT-2000 for developing countries
 - M.1034-1
 - requirements for the radio interface(s)
 - M.1035
 - framework for radio interface(s) and radio sub-system functions
 - M.1036
 - spectrum considerations
 - M.1078
 - security in IMT-2000
 - M.1079
 - speech/voiceband data performance
 - M.1167
 - framework for satellites
 - M.1168
 - framework for management
 - M.1223
 - evaluation of security mechanisms
 - M.1224
 - vocabulary for IMT-2000
 - M.1225
 - evaluation of transmission technologies
 - . . .
- <http://www.itu.int/imt>



Future Wireless Networks

Ubiquitous Communication Among People and Devices



Next-generation Cellular
Wireless Internet Access
Wireless Multimedia
Sensor Networks
Smart Homes/Spaces
Automated Highways
In-Body Networks
All this and more ...

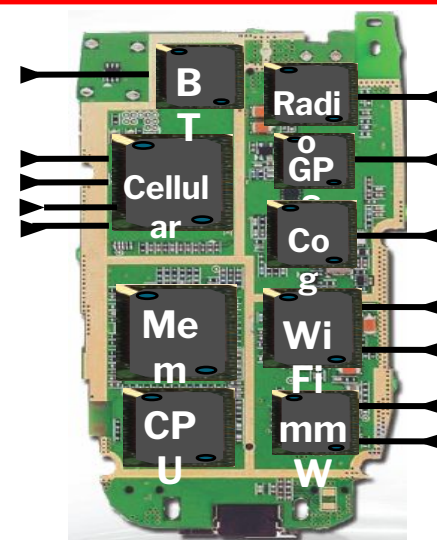
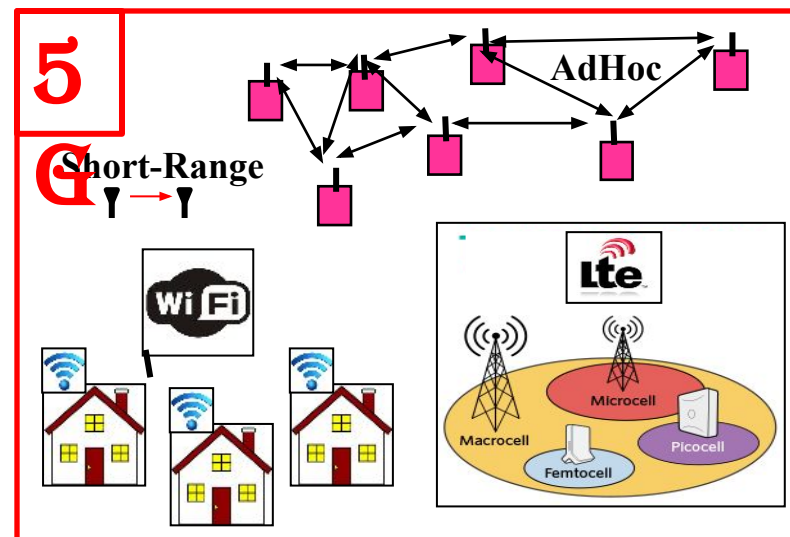
Challenges

- **Network/Radio Challenges**

- Gbps data rates with “no” errors
- Energy efficiency
- Scarce/bifurcated spectrum
- Reliability and coverage
- Heterogeneous networks
- Seamless internetwork handoff

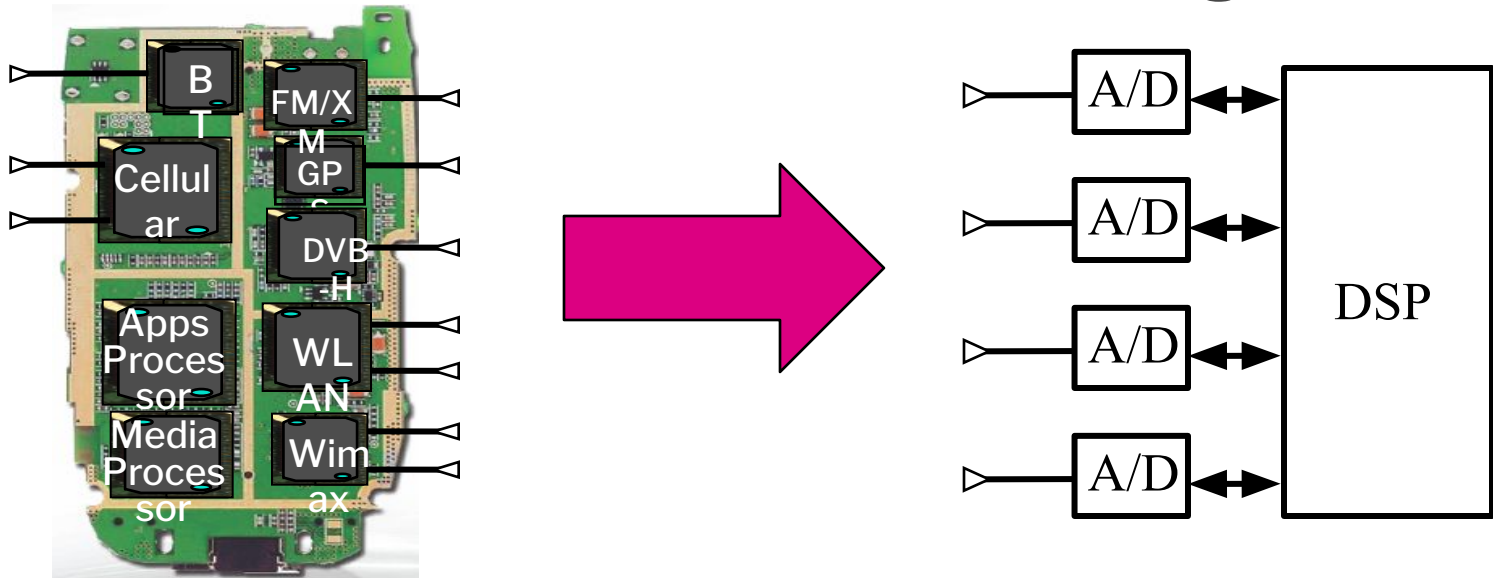
- **Device/SoC (System on a chip) Challenges**

- Performance
- Complexity
- Size, Power, Cost
- High frequencies/mmWave
- Multiple Antennas
- Multiradio Integration
- Coexistence



Software-Defined (SD) Radio:

Is this the solution to the device challenges?

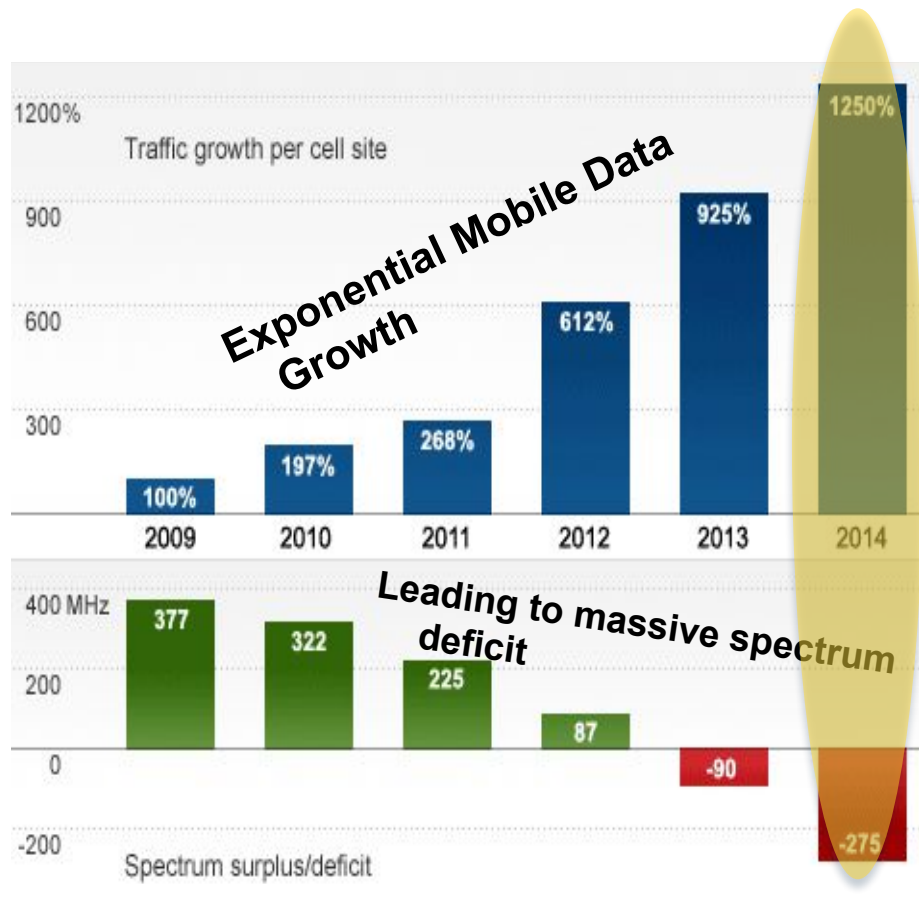


- Wideband antennas and A/Ds span BW of desired signals
- DSP programmed to process desired signal: no specialized HW

Today, this is not cost, size, or power efficient

SubNyquist sampling may help with the A/D and DSP requirements

“Sorry America, your airwaves are full*”



Source: FCC

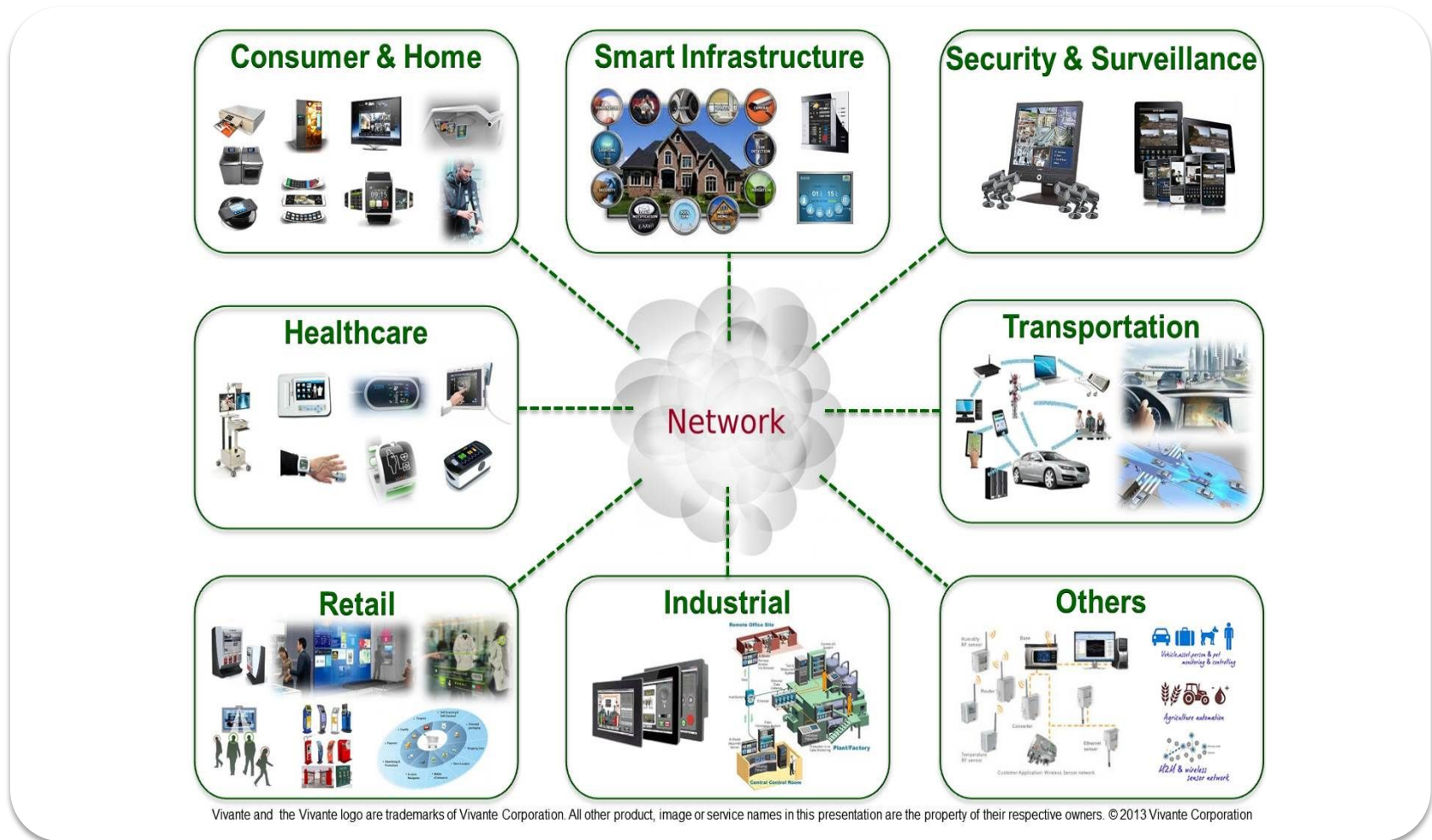
On the Horizon: “The Internet of Things”



50 billion devices by 2020

***CNN MoneyTech – Feb. 2012**

IoT is not (completely) hype



Different requirements than smartphones: **low rates/energy consumption**

Are we at the Shannon limit of the Physical Layer?

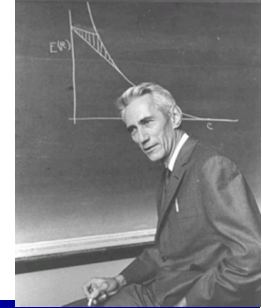
We are at the Shannon Limit

- “The wireless industry has reached the theoretical limit of how fast networks can go” *K. Fitcher, Connected Planet*
- “We’re 99% of the way” to the “barrier known as Shannon’s limit,” *D. Warren, GSM Association Sr. Dir. of Tech.*

Shannon was wrong, there is no limit

- “There is no theoretical maximum to the amount of data that can be carried by a radio channel” *M. Gass, 802.11 Wireless Networks: The Definitive Guide*
- “Effectively unlimited” capacity possible via *personal* cells (pcells). *S. Perlman, Artemis.*

What would Shannon say?



We don't know the Shannon capacity of most wireless channels

- Time-varying channels.
- Channels with interference or relays.
- Cellular systems
- Ad-hoc and sensor networks
- Channels with delay/energy/\$\$\$ constraints.

Shannon theory provides design insights and system performance upper bounds

Current/Emerging Wireless Systems

- **Current:**

- 4G Cellular Systems (LTE-Advanced)
- 4G Wireless LANs/WiFi (802.11ac)
- Satellite Systems
- Bluetooth
- Zigbee
- WiGig

- **Emerging**

- 5G Cellular and WiFi Systems
- mmWave Systems
- Ad/hoc and Cognitive Radio Networks
- Energy-Harvesting Systems

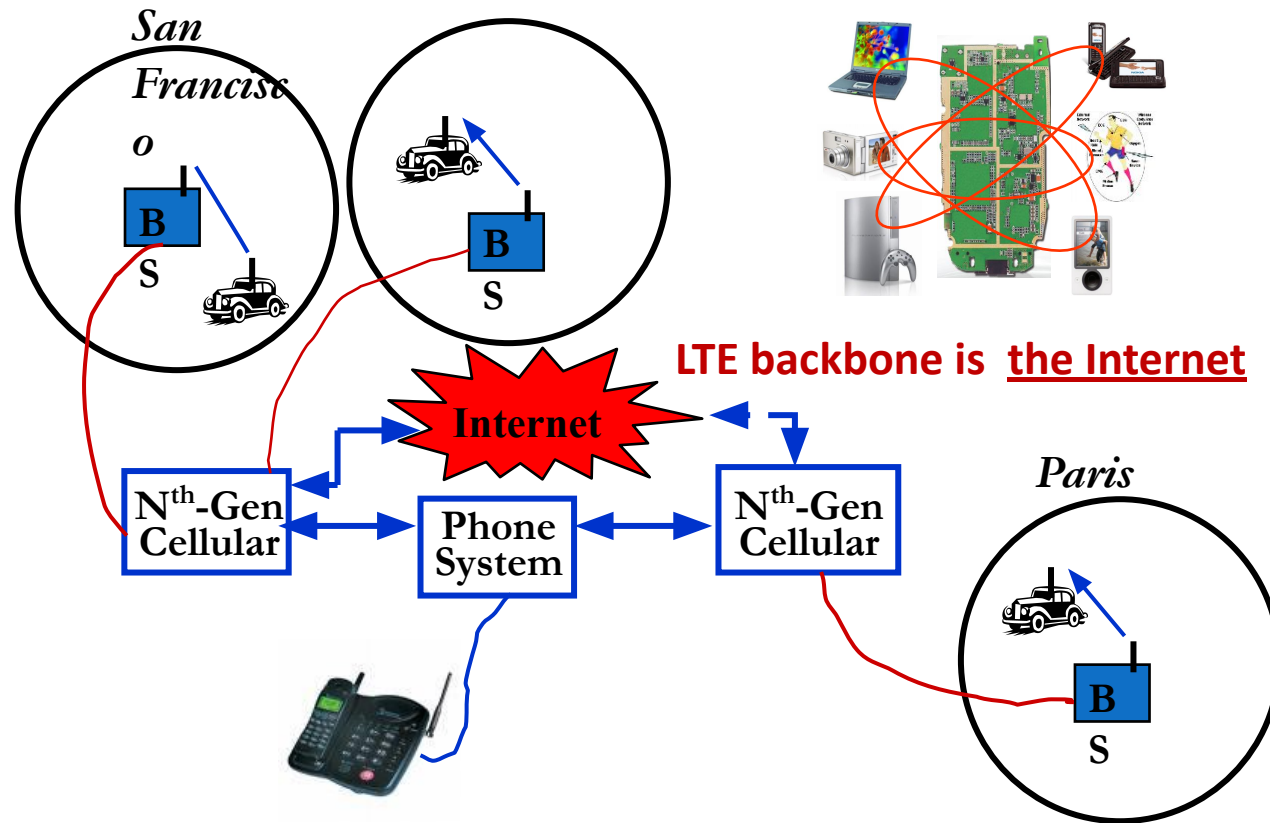
**Much room
For innovation**

4G/LTE Cellular

- Much higher data rates than 3G (50-100 Mbps)
 - 3G systems has 384 Kbps peak rates
- Greater spectral efficiency (bits/s/Hz)
 - More bandwidth, adaptive OFDM-MIMO, reduced interference
- Flexible use of up to 100 MHz of spectrum
 - 10-20 MHz spectrum allocation common
- Low packet latency (<5ms).
- Reduced cost-per-bit (not clear to customers)
- All IP network

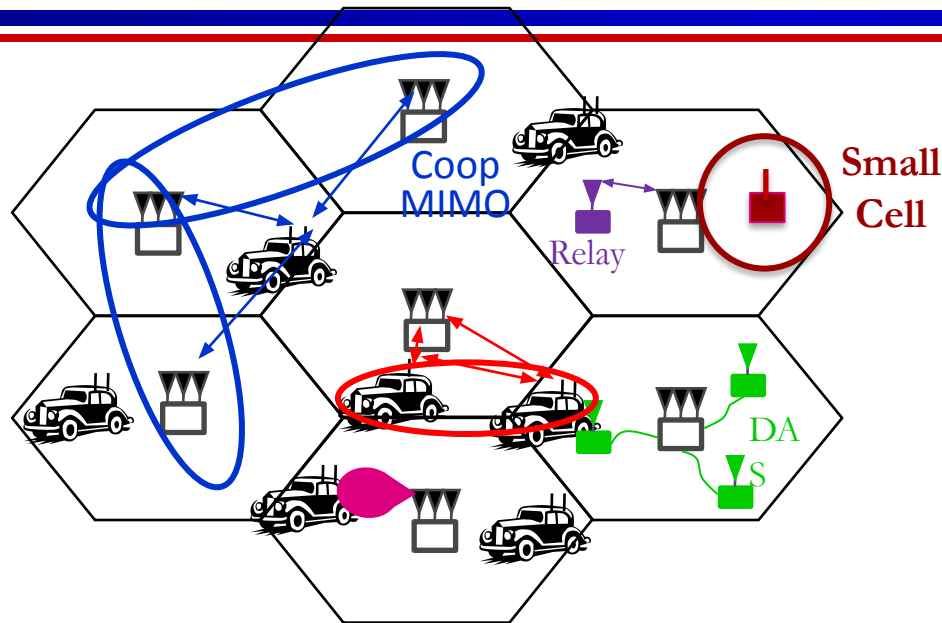
Future Cellular Phones

Burden for this performance is on the backbone network



Much better performance and reliability than today
- *Gbps rates, low latency, 99% coverage indoors and out*

Rethinking “Cells” in Cellular

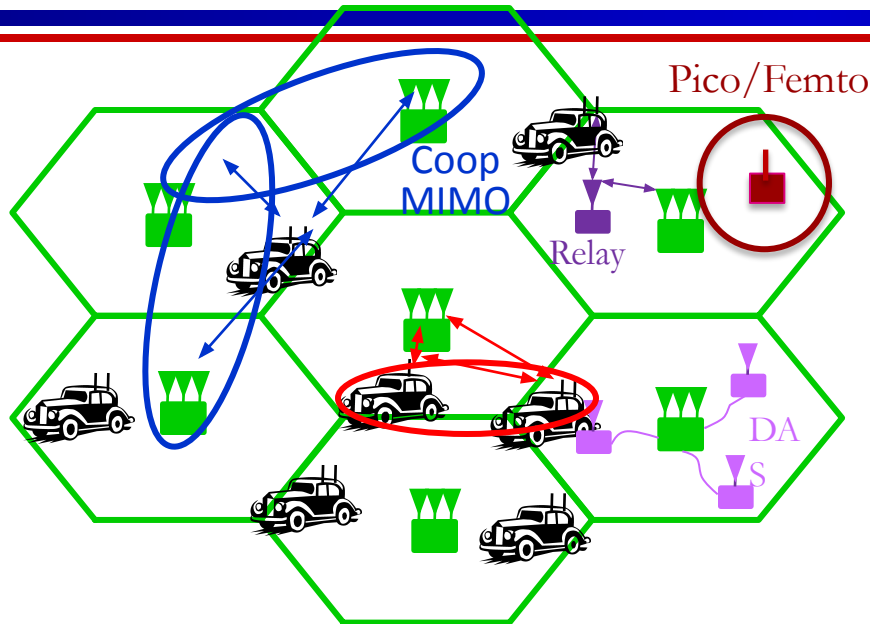


How should cellular systems be designed?

Will gains in practice be big or incremental; in capacity or coverage?

- Traditional cellular design “interference-limited”
 - MIMO/multiuser detection can remove interference
 - Cooperating BSs form a MIMO array: what is a cell?
 - Relays change cell shape and boundaries
 - Distributed antennas move BS towards cell boundary
 - Small cells create a cell within a cell
 - Mobile cooperation via relays, virtual MIMO, network coding.

Green” Cellular Networks



How should cellular systems be redesigned for minimum energy?

Research indicates that significant savings is possible

- Minimize energy at both the mobile and base station via
 - New Infrastructures: cell size, BS placement, DAS, Picos, relays
 - New Protocols: Cell Zooming, Coop MIMO, RRM, Scheduling, Sleeping, Relaying
 - Low-Power (Green) Radios: Radio Architectures, Modulation, coding, MIMO

Wifi Networks

Multimedia Everywhere, Without Wires

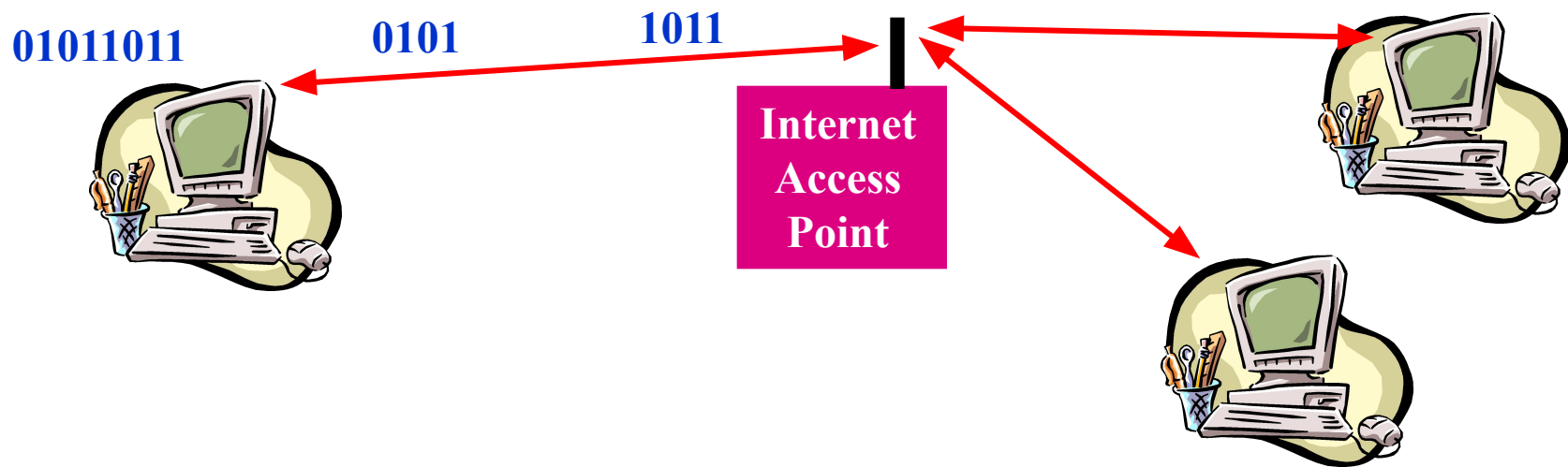


- Streaming video
- Gbps data rates
- High reliability
- Coverage inside and out



Wireless
HDTV
and Gaming

Wireless Local Area Networks (WLANs)



- WLANs connect “local” computers (100m range)
- Breaks data into packets
- Channel access shared (random access + backoff)
- Backbone Internet provides best-effort service
 - Poor performance in some apps (e.g. video)

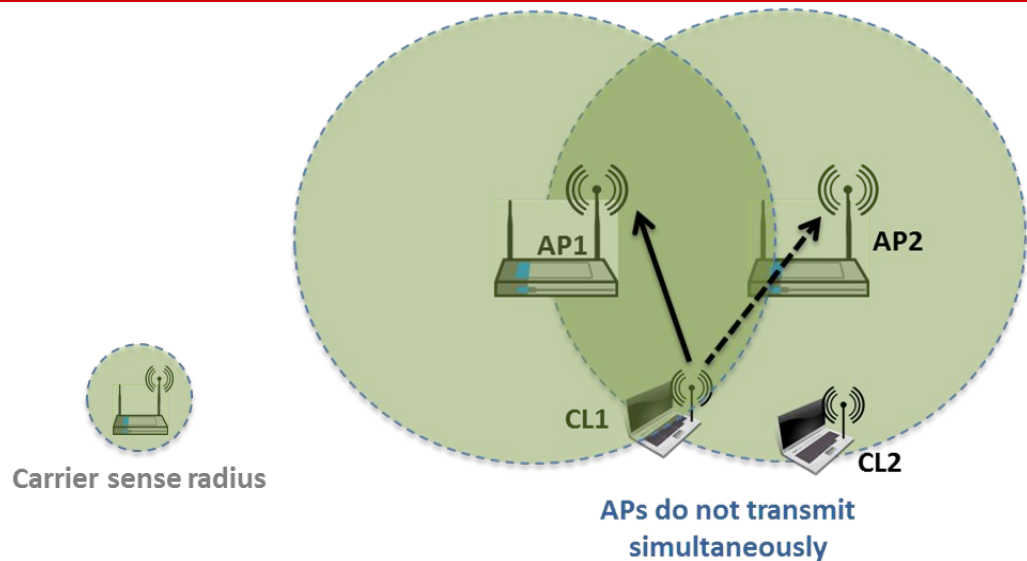
Wireless LAN Standards

- 802.11b (Old – 1990s)
 - Standard for 2.4GHz ISM band (80 MHz)
 - Direct sequence spread spectrum (DSSS)
 - Speeds of 11 Mbps, approx. 500 ft range
 - 802.11a/g (Middle Age– mid-late 1990s)
 - Standard for 5GHz band (300 MHz)/also 2.4GHz
 - OFDM in 20 MHz with adaptive rate/codes
 - Speeds of 54 Mbps, approx. 100-200 ft range
 - 802.11n/ac (Current)
 - Standard in 2.4 GHz and 5 GHz band
 - Adaptive OFDM /MIMO in 20/40/80/160 MHz
 - Antennas: 2-4, up to 8
 - Speeds up to 600Mbps/10 Gbps, approx. 200 ft range
 - Other advances in packetization, antenna use, multiuser MIMO
- Many WLAN cards have (a/b/g/n)

Why does WiFi performance reduce?

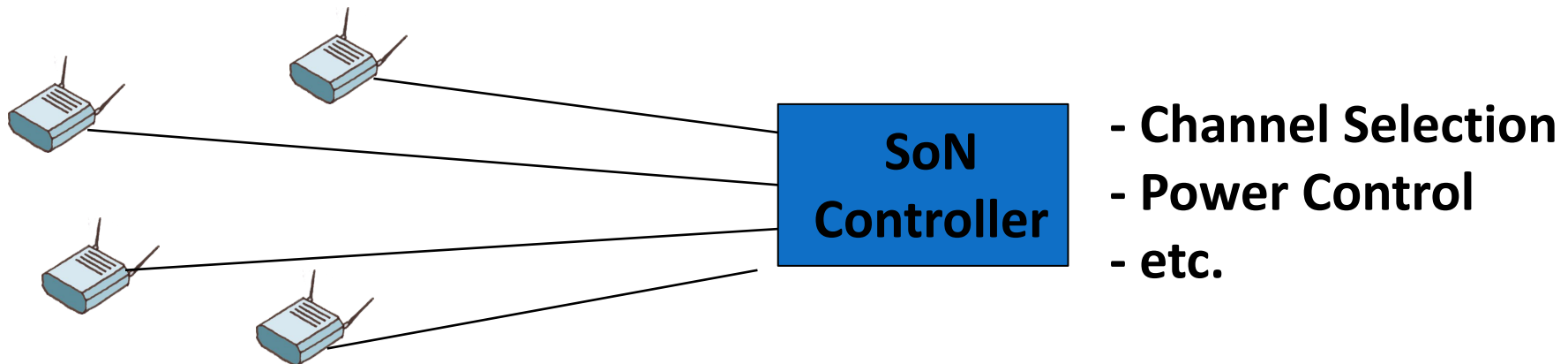
*Carrier Sense Multiple Access:
if another WiFi signal
detected, random backoff*

*Collision Detection: if collision
detected, resend*



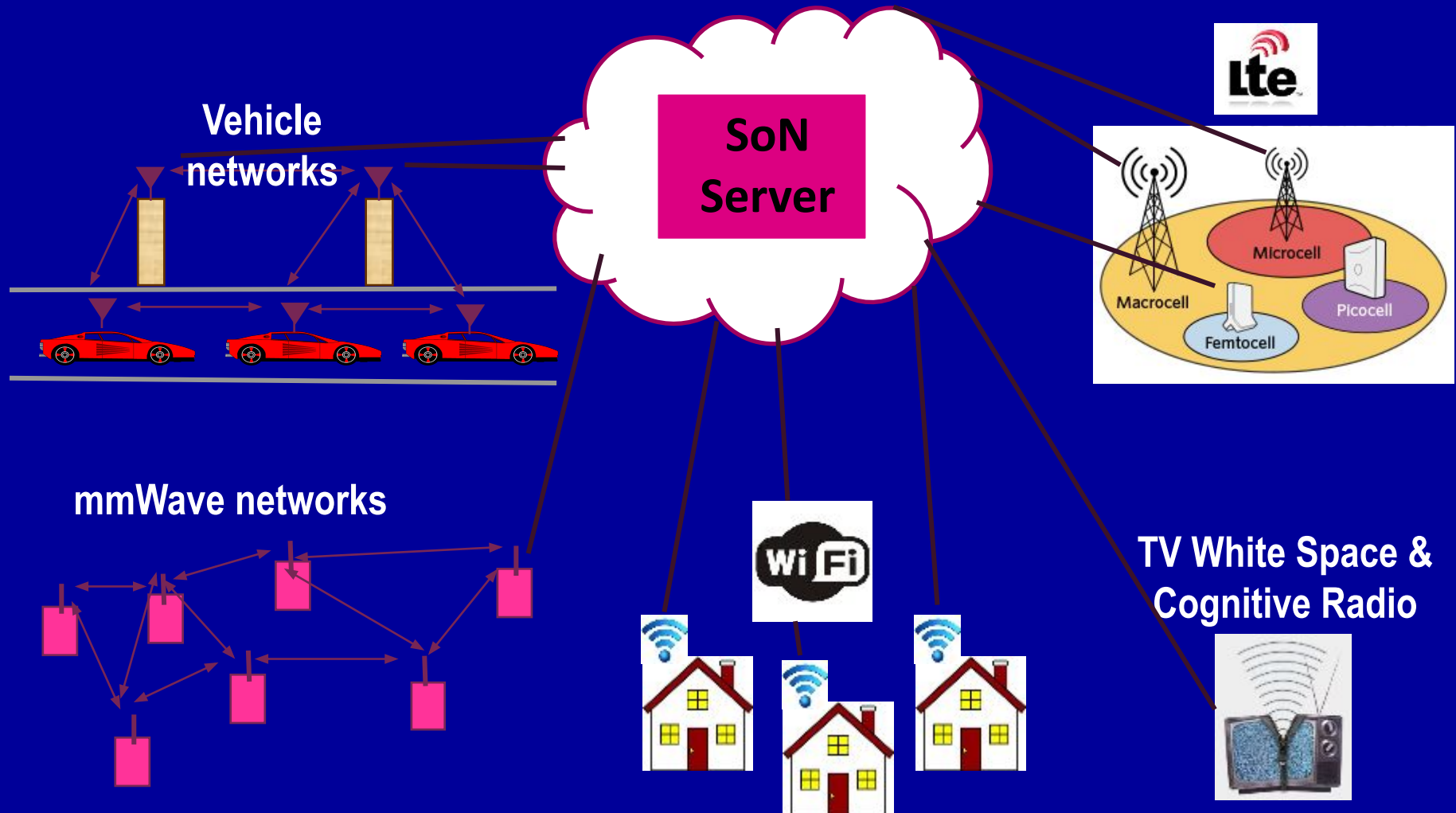
- The WiFi standard lacks good mechanisms to mitigate interference, especially in dense AP deployments
 - Multiple access protocol (CSMA/CD) from 1970s
 - Static channel assignment, power levels, and carrier sensing thresholds
 - In such deployments WiFi systems exhibit poor spectrum reuse and significant contention among APs and clients
 - Result is low throughput and a poor user experience
 - Multiuser MIMO will help each AP, but not interfering APs

Why not use SoN for WiFi?

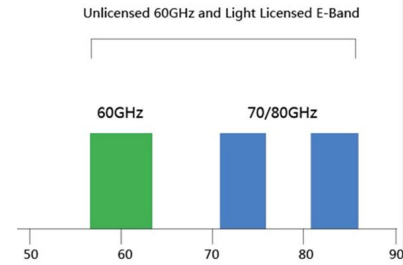


- **SoN-for-WiFi: dynamic self-organization network software to manage of WiFi APs.**
- **Allows for capacity/coverage/interference mitigation tradeoffs.**
- **Also provides network analytics and planning.**

Why not use SON for all Wireless Networks: *Software-Defined Wireless Networks*



WiGig and mmWave



- **WiGig**

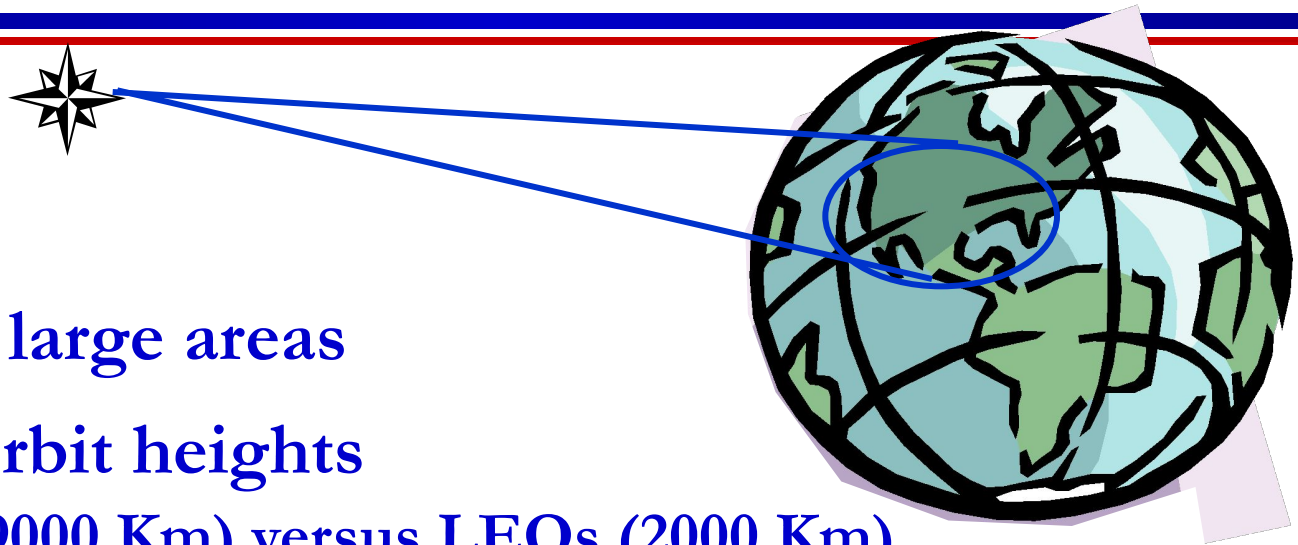
- Standard operating in 60 GHz band
- Data rates of 7-25 Gbps
- Bandwidth of around 10 GHz (unregulated)
- Range of around 10m (can be extended)
- Uses/extends 802.11 MAC Layer
- Applications include PC peripherals, HDTV displays, monitors & projectors. **Not that successful to date.**



- **mmWave**

- 60-80GHz (or higher), massive MIMO, better MAC
- Promises long-range communication w/Gbps data rates
- Hardware, propagation and system design challenges
- Much research on this topic today

Satellite Systems



- Cover very large areas
- Different orbit heights
 - GEOs (39000 Km) versus LEOs (2000 Km)
- Optimized for one-way transmission
 - Radio (XM, Sirius) and movie (SatTV, DVB/S) broadcasts
 - Most two-way systems went bankrupt
- Global Positioning System (GPS) ubiquitous
 - Satellite signals used to pinpoint location
 - Popular in cell phones, PDAs, and navigation devices



Bluetooth®

Bluetooth

- Cable replacement RF technology (low cost)
- Short range (10m, extendable to 100m)
- 2.4 GHz band (crowded)
- 1 Data (700 Kbps) and 3 voice channels, up to 3 Mbps
- Widely supported by telecommunications, PC, and consumer electronics companies
- Few applications beyond cable replacement

IEEE 802.15.4/ZigBee Radios



- Low-rate low-power low-cost secure radio
 - Complementary to WiFi and Bluetooth
- Frequency bands: 784, 868, 915 MHz, 2.4 GHz
- Data rates: 20Kbps, 40Kbps, 250 Kbps
- Range: 10-100m line-of-sight
- Support for large mesh networking or star clusters
- Support for low latency devices
- CSMA-CA channel access
- Applications: light switches, electricity meters, traffic management, and other low-power sensors.

Spectrum Regulation

- Spectrum a scarce public resource, hence allocated
- Spectral allocation in US controlled by FCC (commercial) or OSM (defense)
- FCC auctions spectral blocks for set applications.
- Some spectrum set aside for universal use
- Worldwide spectrum controlled by ITU-R
- Regulation is a necessary evil.

Innovations in regulation being considered worldwide
in multiple cognitive radio paradigms

Standards

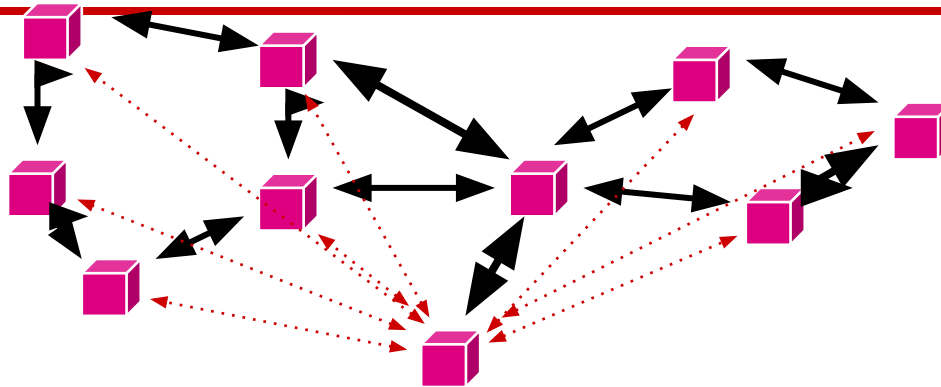
- Interacting systems require standardization
- Companies want their systems adopted as standard
 - Alternatively try for de-facto standards
- Standards determined by TIA/CTIA in US
 - IEEE standards often adopted
 - Process fraught with inefficiencies and conflicts
- Worldwide standards determined by ITU-T
 - In Europe, ETSI is equivalent of IEEE

Standards for current systems are summarized in Appendix D.

Emerging Systems

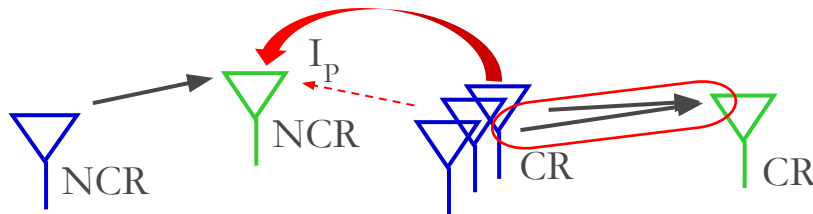
- Ad hoc/mesh wireless networks
- Cognitive radio networks
- Wireless sensor networks
- Energy-harvesting radios
- Distributed control networks
- Applications of Communications in Health, Bio-medicine, and Neuroscience

Ad-Hoc Networks

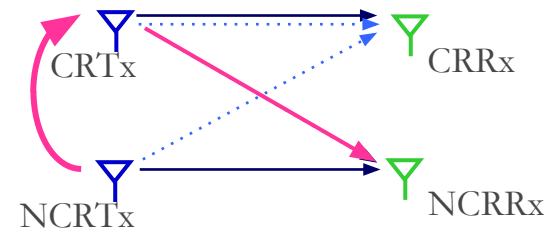


- Peer-to-peer communications
 - No backbone infrastructure or centralized control
- Routing can be multihop.
- Topology is dynamic.
- Fully connected with different link SINRs
- Open questions
 - Fundamental capacity region
 - Resource allocation (power, rate, spectrum, etc.)
 - Routing

Cognitive Radios



MIMO Cognitive Underlay

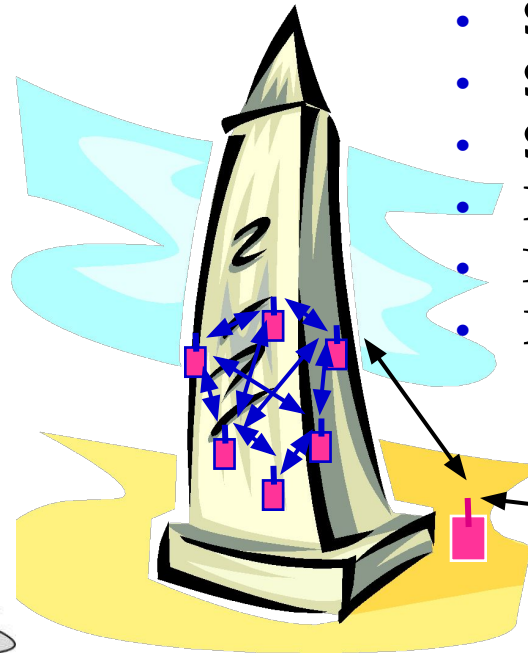
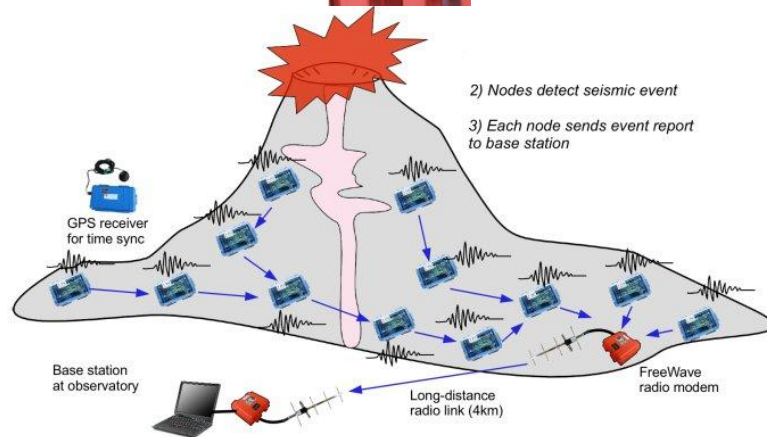


Cognitive Overlay

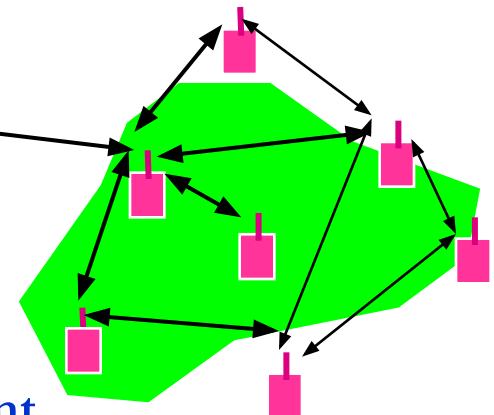
- Cognitive radios support new users in existing crowded spectrum without degrading licensed users
 - Utilize advanced communication and DSP techniques
 - Coupled with novel spectrum allocation policies
- Multiple paradigms
 - (MIMO) Underlay (interference below a threshold)
 - Interweave finds/uses unused time/freq/space slots
 - Overlay (overhears/relays primary message while cancelling interference it causes to cognitive receiver)

Wireless Sensor Networks

Data Collection and Distributed Control



- Smart homes/buildings
- Smart structures
- Search and rescue
- Homeland security
- Event detection
- Battlefield surveillance



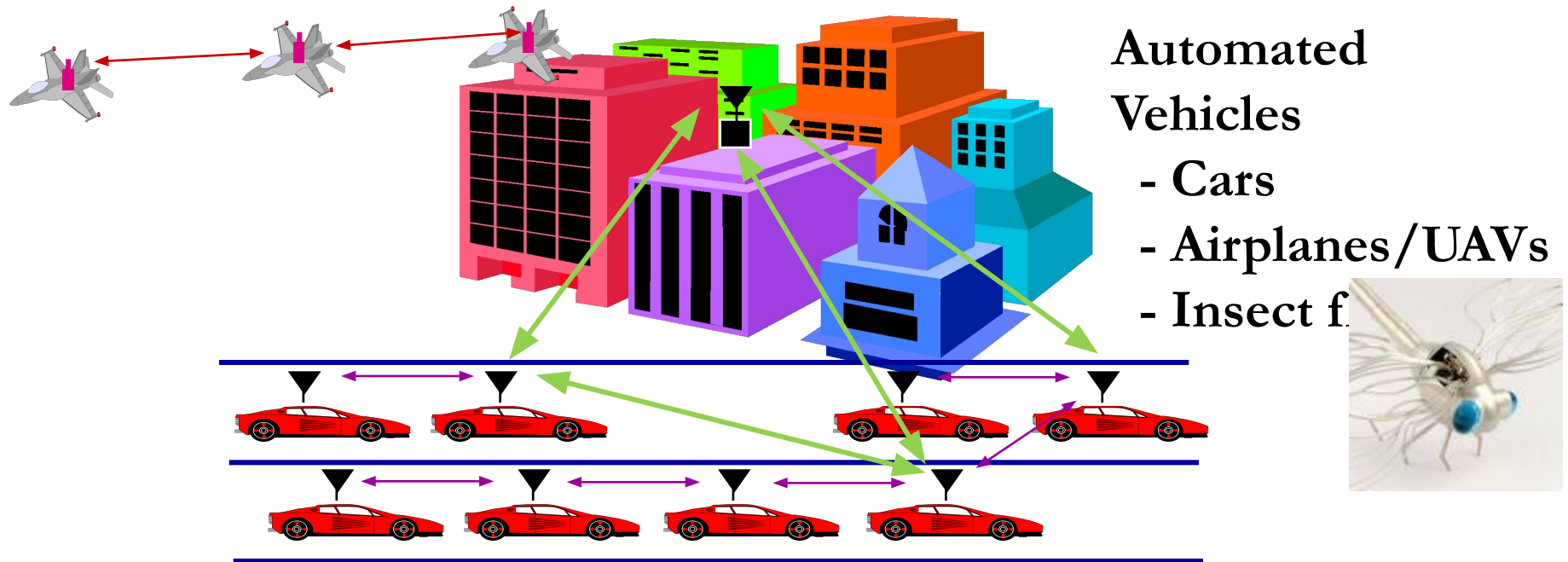
- Energy (transmit and processing) is the driving constraint
- Data flows to centralized location (joint compression)
- Low per-node rates but tens to thousands of nodes
- Intelligence is in the network rather than in the devices

Energy-Harvesting Radios



- How should radios be powered?
 - Batteries and traditional charging mechanisms
 - Wireless-power transfer (poorly understood)
 - By harvesting energy from the environment
- Radios with intermittent random energy arrivals
 - New communication design principles needed (Ozgur)

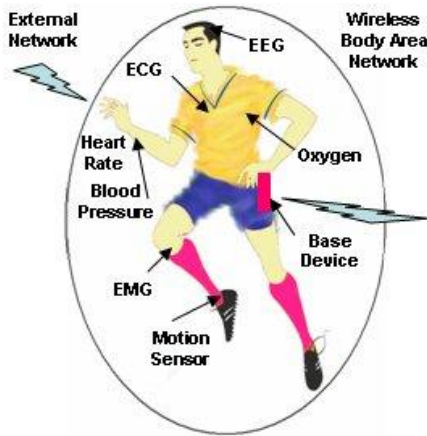
Distributed Control over Wireless



Interdisciplinary design approach

- Control requires **fast**, **accurate**, and **reliable** feedback.
- Wireless networks introduce **delay** and **loss**
- Need reliable networks and robust controllers
- Mostly open problems : *Many design challenges*

Applications in Health, Biomedicine and Neuroscience



Body-Area
Networks



Neuro/Bioscience

- EKG signal reception/modeling
- Brain information theory
- Nerve network (re)configuration
- Implants to monitor/generate signals
- In-brain sensor networks
- SP/Comm applied to bioscience

Doctor-on-a-chip

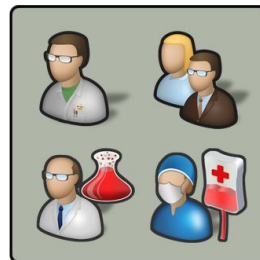


Wireless
Network

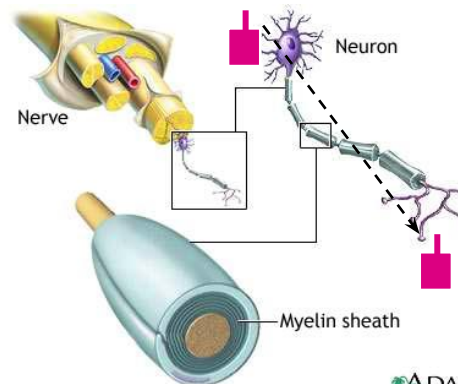


EHR

Telemedicine



Specialist Network



Recovery from Nerve Damage

Main Points

- The wireless vision encompasses many exciting applications
- Technical challenges transcend all system design layers
- 5G networks must support higher performance for some users and extreme energy efficiency for others
- Cloud-based software to dynamically control and optimize wireless networks needed (SDWN)
- Innovative wireless design needed for 5G cellular/WiFi, mmWave systems, massive MIMO, and IoT connectivity
- Standards and spectral allocation heavily impact the evolution of wireless technology